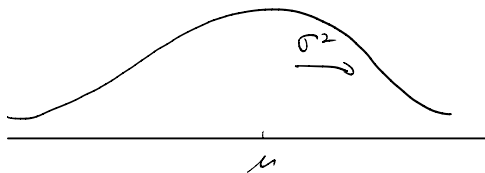


Review

Normal Distribution $X \sim N(\mu, \sigma^2)$ notation



Standard Normal $Z \sim N(\mu=0, \sigma^2=1)$

table (text + Quercus)

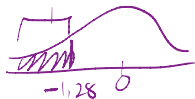


area to right
of the

In R

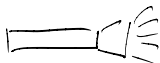
`pnorm(z)` area to the left of standard normal by default

`pnorm(-1.28)`



`pnorm(+2.56)`





Empirical Rule (68-95-99.7 Rule)

For (approx) bell shaped distributions

Approx 68% of observations within 1 st. dev of the mean

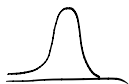
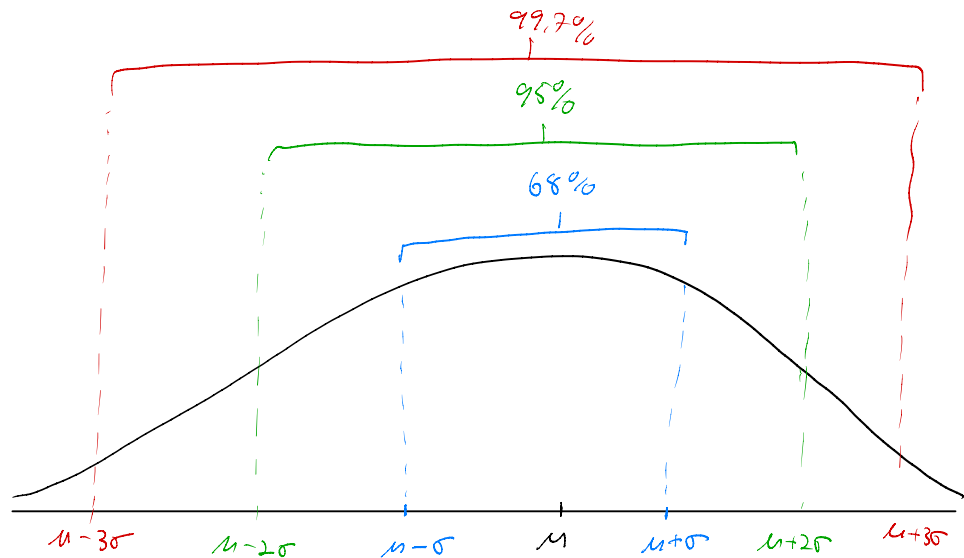
$$\mu \pm 1\sigma$$

Approx 95% of observations within 2 st. dev of the mean

$$\mu \pm 2\sigma$$

Approx 99.7% of observations within 3 st. dev of the mean

$$\mu \pm 3\sigma$$



STA258H5

University of Toronto Mississauga

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Winter 2023

THE CENTRAL LIMIT THEOREM (CLT)

Theorem 7.5

Proof : left for self study

Let Y and $Y_1, Y_2, Y_3, \dots$ be random variables with moment-generating functions $M(t)$ and $M_1(t), M_2(t), \dots$, respectively. If

$$\lim_{n \rightarrow \infty} M_n(t) = M(t) \quad \text{for all real } t,$$

then the distribution function of Y_n converges to the distribution function of Y as $n \rightarrow \infty$.

Maclaurin Series

A Maclaurin series is a Taylor series expansion of a function about 0,

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \dots + \frac{f^{(n)}(0)x^n}{n!} + \dots$$

Maclaurin series are named after the Scottish mathematician Colin Maclaurin.

Useful properties of MGFs

- $M_Y(0) = E(e^{0Y}) = E(1) = 1.$
- $M'_Y(0) = E(Y).$
- $M''_Y(0) = E(Y^2).$
- $M_{aY}(t) = E(e^{t(aY)}) = E(e^{(at)Y}) = M_Y(at),$ where a is a constant.

Sampling Distribution

calculate from data

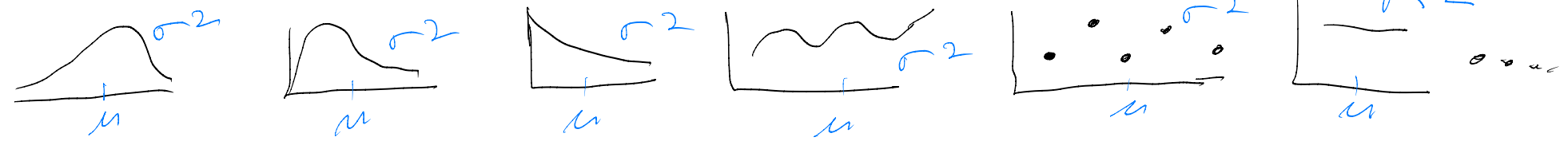


Distribution of a statistic

e.g. the distribution of sample mean $\bar{x}$

Central Limit Theorem

We have a population which has mean μ and St. dev σ



repeatedly take many samples where each sample is of size n

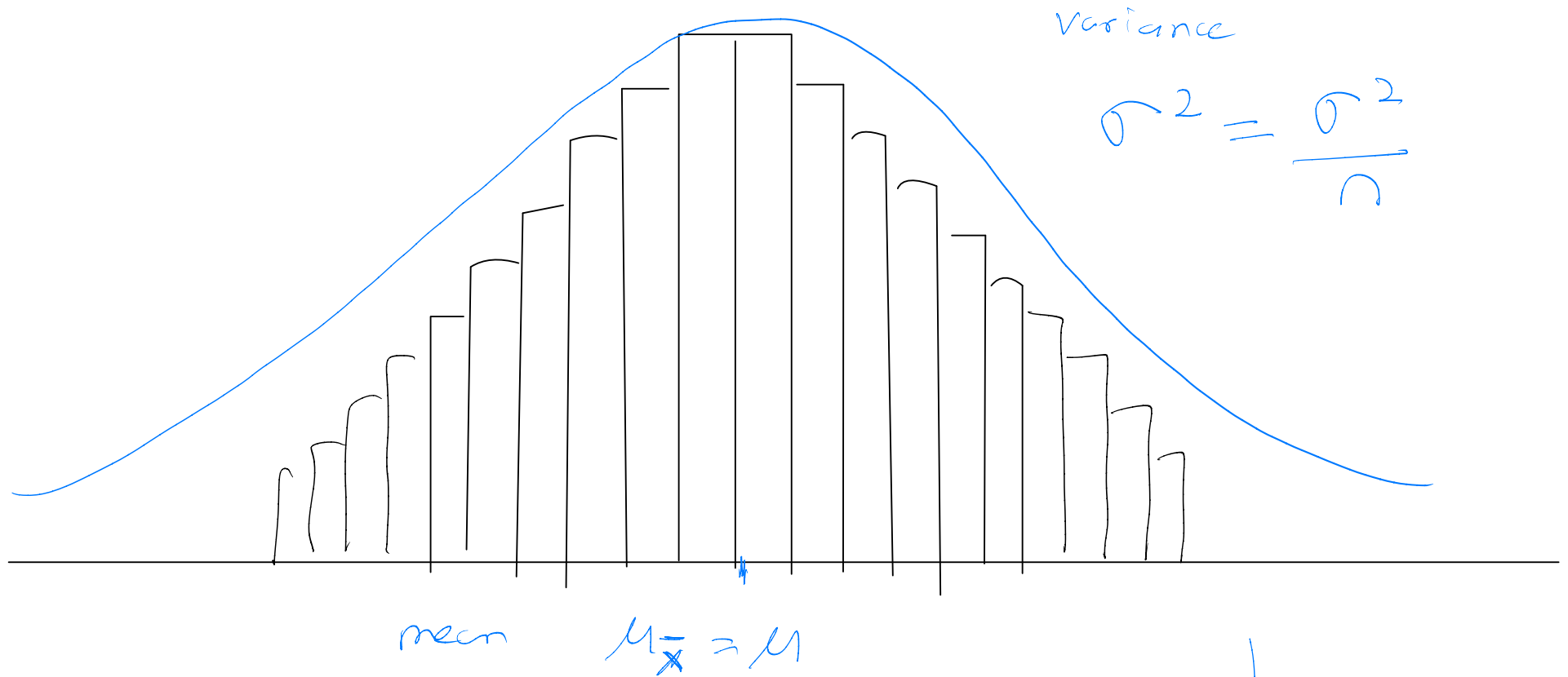
Sample 1 Sample 2 Sample 3 ... Sample 10,000

$n \begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$ $n \begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$ $n \begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$... $n \begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$

$\bar{X}_1$ $\bar{X}_2$ $\bar{X}_3$... $\bar{X}_{10,000}$

large collection
of $\bar{X}$'s

create distribution for
this collection of $\bar{X}$'s



The sampling distribution of $\bar{X}$ is

$$\bar{X} \sim N\left(\mu_{\bar{x}} = \mu, \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}\right)$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Note:
large

$$n \geq 30$$

The larger the sample size, the more normal the sampling distribution

Follow up result for totals

$$\text{var}(aX) = a^2 \text{var}(X)$$

$$\text{var}(n\bar{X}) = n^2 \text{var}(\bar{X}) = n^2 \cdot \frac{\sigma^2}{n} = n\sigma^2$$

$$\bar{X} = \frac{(\sum x_i)_{\text{total}}}{n}$$

$$n\bar{X} = \sum x_i$$

$$T = \sum x_i \sim N\left(\mu_{\text{Total}} = n\mu, \sigma_{\text{Total}}^2 = n\sigma^2\right)$$

Notes on proof (Self Study)

↳ obtained using Taylor Series + MGF

Central Limit Theorem

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with $E(X_i) = \mu$ and $V(X_i) = \sigma^2 < \infty$. Define

$$U_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

Then the distribution function of U_n converges to the standard Normal distribution function as $n \rightarrow \infty$. That is,

$$\lim_{n \rightarrow \infty} P(U_n \leq u) = \int_{-\infty}^u \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \text{ for all } u.$$

Let $Z_i = \frac{X_i - \mu}{\sigma}$. Note that $E(Z_i) = 0$ and $V(Z_i) = 1$. Let us rewrite U_n

$$\sqrt{n} \left(\frac{\bar{X} - \mu}{\sigma} \right) = \sqrt{n} \left(\frac{\sum_{i=1}^n X_i - n\mu}{n\sigma} \right) = \frac{1}{\sqrt{n}} \left(\frac{\sum_{i=1}^n X_i - n\mu}{\sigma} \right)$$

$$U_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right) = \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i.$$

Since the mgf of the sum of independent random variables is the product of their individual mgfs, if $M_{Z_i}(t)$ denotes the mgf of each random variable Z_i

$$M_{\sum_{i=1}^n Z_i}(t) = [M_{Z_1}(t)]^n$$

and

$$M_{U_n} = M_{\sum_{i=1}^n Z_i}(t/\sqrt{n}) = [M_{Z_1}(t/\sqrt{n})]^n.$$

Recall that $M_{Z_i}(0) = 1$, $M'_{Z_i}(0) = E(Z_i) = 0$, and $M''_{Z_i}(0) = E(Z_i^2) = V(Z_i^2) = 1$.

Now, let us write the Taylor's series of $M_{Z_i}(t)$ at 0

$$M_{Z_i}(t) = M_{Z_i}(0) + tM'_{Z_i}(0) + \frac{t^2}{2!}M''_{Z_i}(0) + \frac{t^3}{3!}M'''_{Z_i}(0) + \frac{t^4}{4!}M''''_{Z_i}(0) + \dots$$

$$M_{Z_i}(t) = 1 + \frac{t^2}{2} + \frac{t^3}{3!}M'''_{Z_i}(0) + \frac{t^4}{4!}M''''_{Z_i}(0) + \dots$$

$$M_{U_n}(t) = [M_{Z_1}(t/\sqrt{n})]^n = \left[1 + \frac{t^2}{2n} + \frac{t^3}{3!n^{3/2}}M'''_{Z_i}(0) + \frac{t^4}{4!n^2}M''''_{Z_i}(0) + \dots\right]^n$$

Recall that if

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

$$\lim_{n \rightarrow \infty} b_n = b \quad \lim_{n \rightarrow \infty} \left(1 + \frac{b_n}{n}\right)^n = e^b$$

But

$$\lim_{n \rightarrow \infty} \left[\frac{t^2}{2} + \underbrace{\frac{t^3}{3!n^{1/2}} M_{Z_i}'''(0) + \frac{t^4}{4!n} M_{Z_i}''''(0) + \dots}_{\text{terms that go to 0}} \right] = \frac{t^2}{2}$$

Therefore,

$$\lim_{n \rightarrow \infty} M_{U_n}(t) = \exp\left(\frac{t^2}{2}\right)$$

which is the moment-generating function for a standard Normal random variable. Applying Theorem 7.5 we conclude that U_n has a distribution function that converges to the distribution function of the standard Normal random variable.

Using the CLT

sample mean

Two approximate distributions:

① $\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$

② $T = \sum_{i=1}^n X_i \approx N(n\mu, n\sigma^2)$

total

Example

The number of accidents per week at a hazardous intersection varies with mean 2.2 and standard deviation 1.4. This distribution takes only whole-number values, so it is certainly not Normal.

a) Let $\bar{x}$ be the mean number of accidents per week at the intersection during a year (52 weeks). What is the approximate distribution of $\bar{x}$ according to the Central Limit Theorem?

b) What is the approximate probability that $\bar{x}$ is less than 2?

c) What is the approximate probability that there are fewer than 100 accidents at the intersection in a year?

52 weeks

< 100 total

Example (Slide 12)

$\mu = 2.2$ $\sigma = 1.4$ not normal

$\hookrightarrow \sigma^2 = 1.4^2$

a) $n = 52$

mean

variance

By CLT $\bar{X} \sim N\left(\mu_{\bar{X}} = \mu = 2.2, \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n} = \frac{1.4^2}{52} = 0.03767\right)$

$$\bar{X} \sim N(\mu_{\bar{X}} = 2.2, \sigma_{\bar{X}}^2 = 0.03767)$$

$\hookrightarrow \sigma_{\bar{X}} = \sqrt{\sigma_{\bar{X}}^2} = 0.1941$

b) Transformation $Z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}}$

mean of $\bar{X}$

$$\sigma_{\bar{X}} = \sqrt{\sigma_{\bar{X}}^2}$$

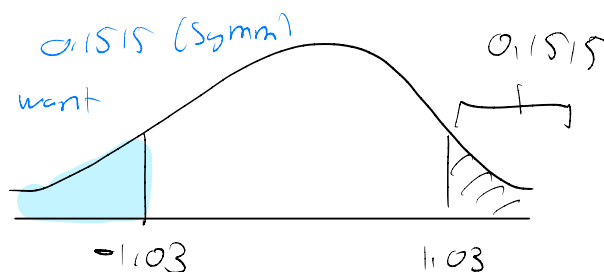
$$= \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}$$

$$P(\bar{X} < 2) = P\left(\frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} < \frac{2 - \mu_{\bar{X}}}{\sigma_{\bar{X}}}\right)$$

$$= P\left(Z < \frac{2 - 2.2}{0.1941}\right)$$

$$= P(Z < -1.03)$$

$$= 0.1515$$



c) Prob less than 100 total accidents

$$T = \sum x_i \sim N\left(\mu_{\text{Total}} = n\mu = (52)(2.2), \sigma_{\text{Total}}^2 = n\sigma^2 = (52)(1.4)^2 = 101.92\right)$$

$$\sim N(\mu_{\text{Total}} = 114.4, \sigma_{\text{Total}}^2 = 101.92)$$

$\hookrightarrow \sigma_{\text{Total}} = \sqrt{101.92} = 10.096$

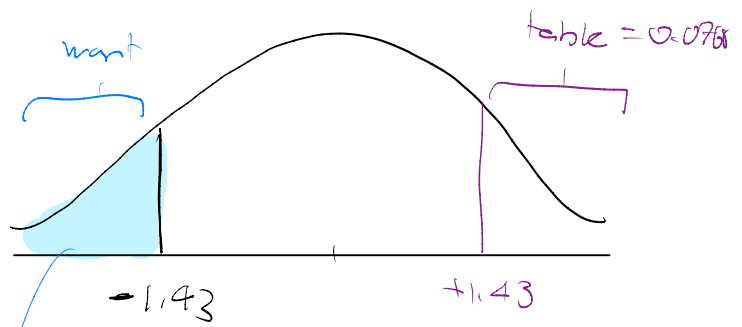
$$P(T < 100)$$

$$= P\left(\frac{T - \mu_{\text{Total}}}{\sigma_{\text{Total}}} < \frac{100 - \mu_{\text{Total}}}{\sigma_{\text{Total}}}\right)$$

$$= P\left(Z < \frac{100 - 114.4}{10.0\%}\right)$$

$$= P(Z < -1.43)$$

$$= 0.0768$$



In R
pnorm(-1.43)

Solution

a) By the Central Limit Theorem, $\bar{X}$ is roughly Normal with mean $\mu^* = 2.2$ and standard deviation $\sigma^* = \sigma/\sqrt{n} = 1.4/\sqrt{52} = 0.1941$.

$$\begin{aligned} \text{b) } P(\bar{X} < 2) &= P\left(\frac{\bar{X} - \mu^*}{\sigma^*} < \frac{2 - 2.2}{0.1941}\right) \\ &\approx P(Z < -1.0303) = 0.1515 \end{aligned}$$

Solution

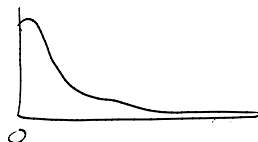
Let X_i be the number of accidents during week i .

$$\begin{aligned} \text{c) } P(\text{Total} < 100) &= P\left(\sum_{i=1}^{52} X_i < 100\right) \\ &= P\left(\frac{\sum_{i=1}^{52} X_i}{52} < \frac{100}{52}\right) \\ &= P(\bar{X} < 1.9230) \\ &\approx P(Z < -1.4270) = 0.0768 \end{aligned}$$

More on insurance

$$\mu = 75$$

$$\sigma = 300$$



An insurance company knows that in the entire population of millions of apartment owners, the mean annual loss from damage is $\mu = 75$ and the standard deviation of the loss is $\sigma = 300$. The distribution of losses is strongly right-skewed: most policies have \$0 loss, but a few have large losses. If the company sells 10,000 policies, can it safely base its rates on the assumption that its average loss will be no greater than \$85?

$\bar{x}$

$\leq$

85

Example 15

heavily
right skew

$$\mu = 75, \quad \sigma = 300$$

$$n = 10,000$$

will $\bar{X} \leq 85$

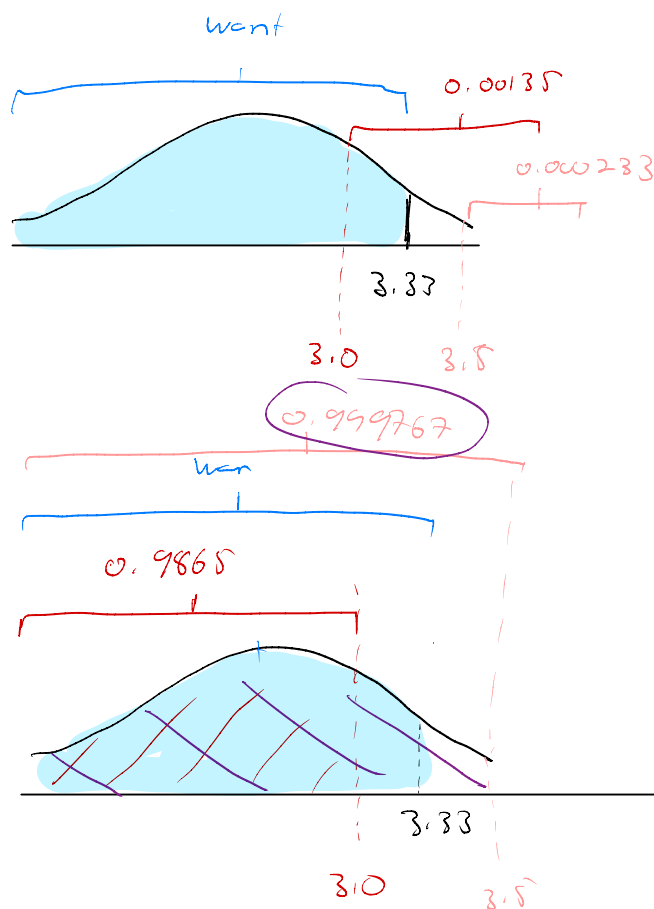
By CLT (10,000 v. large)

$$\bar{X} \sim N\left(\mu_{\bar{X}} = \mu = 75, \quad \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n} = \frac{300^2}{10,000} = 9\right)$$

$$\hookrightarrow \sigma_{\bar{X}} = \sqrt{\sigma_{\bar{X}}^2} = 3$$

$$\begin{aligned} P(\bar{X} < 85) \\ &= P\left(Z < \frac{85 - 75}{3}\right) \\ &= P(Z < 3.33) \end{aligned}$$

$$0.9865 < P(Z < 3.33) < 0.999$$



B

$$\text{pnorm}(3.33)$$

The Central Limit Theorem says that, in spite of the skewness of the population distribution, the average loss among 10,000 policies will be approximately $N(75; \frac{300}{\sqrt{10000}}) = N(75; 3)$. Now

$$P(\bar{X} > 85) \approx P\left(Z > \frac{85 - 75}{3}\right) = P(Z > 3.33) = 1 - 0.9996 = 0.0004$$

We can be about 99.96% certain that average losses will not exceed \$85 per policy.

A freight elevator can transport a maximum of 9800 kg. Suppose a load of cargo containing 49 boxes must be transported via the elevator. Experience has shown that the weight of boxes of this type of cargo follows a distribution with mean 205 kg and standard deviation 15 kg. Based on this information, what is the probability that all 49 boxes can be safely loaded onto the freight elevator and transported?

Example

A manufacturer of automobile batteries claims that the distribution of the lengths of life of its best battery has a mean of 54 months and a standard deviation of 6 months. Suppose a consumer group decides to check the claim by purchasing a sample of 50 of the batteries and subjecting them to tests that estimate the battery's life.

- a) Assuming that the manufacturer's claim is true, describe the sampling distribution of the mean lifetime of a sample of 50 batteries.
- b) Assuming that the manufacturer's claim is true, what is the probability that the consumer group's sample has a mean life of 52 or fewer months?

Solution a)

We can use the Central Limit Theorem to deduce that the sampling distribution for a sample mean lifetime of 50 batteries is approximately Normally distributed. Furthermore, the mean of this sampling distribution ($\mu_{\bar{X}}$) is 54 months according to the manufacturer's claim. Finally, the standard deviation of the sampling distribution is given by

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{6}{\sqrt{50}} = 0.8485 \text{ month}$$

Solution b)

If the manufacturer's claim is true, the probability that the consumer group observes a mean battery life of 52 or fewer months for its sample of 50 batteries is given by

$P(\bar{X} \leq 52)$, where $\bar{X}$ is Normally distributed, $\mu_{\bar{X}} = 54$ and $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{6}{\sqrt{50}} = 0.8485$. Hence,

$$\begin{aligned} P(\bar{X} \leq 52) &= P\left(\frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} \leq \frac{52 - 54}{0.8485}\right) \\ &\approx P(Z \leq -2.35710076605775) \\ &\approx P(Z \leq -2.36) \end{aligned}$$

$$\begin{aligned} &\text{(from Table)} \\ &= 0.0091 \end{aligned}$$